

MUHAMMAD UMAR FAROOQ

AI/ML ENGINEER

momerfarooq223@gmail.com | +92 328 6403551 | Lahore, Pakistan

[GitHub](#) | [LinkedIn](#) | [Portfolio](#) | [LeetCode](#) | [Hugging Face](#) | [Kaggle](#)

PROFILE

BS Artificial Intelligence student at UMT (3.85 CGPA, **Dean's & Rector's Honor list**) and **1st-place winner of the Gemma Fine-Tuning Hackathon**. DataCamp certified AI Engineer Associate with **16+ credentials** across DataCamp, Anthropic, OpenAI, and Google. Specializing in autonomous multi-agent architectures (MCP, Claude Code, LangGraph), LLM fine-tuning (QLoRA), and production AI pipelines with **27+** shipped projects and **14+** agents.

EDUCATION

University of Management & Technology (UMT), Lahore

Oct 2023 - Present

BS Artificial Intelligence | 7th Semester | CGPA: 3.85/4.00

- 70% Merit Scholarship awarded based on academic performance
- **Honors:** Rector's Award (Semester 4.0 SGPA) | Dean's Award (Top 10% AI Dept SGPA)

Punjab Group of Colleges (PGC), Lahore

Sep 2021 - Jun 2023

FSc Pre-Medical | 970/1100 (88.2%) | 50% Merit Scholarship

PROFESSIONAL DEVELOPMENT

Panaversity - Agentic AI Program

Jan 2026 – Present

Learning: Agentic AI, Claude Code, Tool Integration, MCP | Focus: foundational agentic patterns and advanced agent design

EXPERIENCE

UMT - Peer Tutoring & Online Instruction

Feb 2025 - Sep 2025

Academic Tutor | Lahore, Pakistan

- Tutored university peers across AI and programming coursework, improving comprehension through adaptive pedagogy.
- Designed and delivered an online technical summer course (Jul–Sep 2025), conducting live coding sessions and mentoring projects.
- Strengthened technical communication and mentoring skills by translating complex algorithms into hands-on workflows.

Customer Services & Operations Roles

Nov 2023 – Sep 2024

Remote & Lahore, Pakistan

- Shopify Store (Nov 2023 - Jan 2024): resolved customer inquiries, handled orders/returns, and managed shipment updates.
- SMZ Dispatch (Feb 2024 - Apr 2024): coordinated with US-based clients, improving cross-cultural communication and reliability.
- Grasp Solutions (Jun 2024 - Sep 2024): managed high-volume client interactions and taxi-dispatch coordination under time pressure.

Techo hub — Agentic AI Developer Intern

June 2026 - Present

Hybrid & Lahore, Pakistan

- Architected domain-aware decision-intelligence systems with multi-tenant vector retrieval, citation grounding, and automated continuous monitoring.
- Engineered local-first multimodal desktop agents combining on-device LLM inference with cloud fallbacks and real-time screen cognition.
- Developed applied computer vision and mathematical modeling pipelines for industrial automation and closed-loop process optimization.

PROJECTS

Gemma Fine-Tuning - 1st Place Hackathon Winner

github.com/omerfarooq223/gemma-claim-verification

Gemma 4 12B | 4-bit NF4 QLoRA | PEFT | PyTorch | NLP

- Fine-tuned a 12B Gemma model using 4-bit NF4 QLoRA on 1,085 audited and contrastive training examples.
- Achieved 94.40% accuracy and 94.38% Macro-F1 on the 500-example supervised final evaluation.
- Used semantic data auditing, targeted contrastive examples, deterministic inference and robustness evaluation to select the final checkpoint.

Personal AI Employee

github.com/omerfarooq223/personal-ai-employee

Claude Code | Flask | Gmail API | Playwright | MCP-style tooling

- Built an autonomous human-in-the-loop assistant that monitors Gmail, reasons over incoming tasks, and prepares approval-ready actions.
- Integrated Playwright headless browser automation to draft, format, and auto-publish contextual content to LinkedIn upon dashboard approval.

AutoGrader Agent

github.com/omerfarooq223/AutoGrader-Agent

Python | Groq/Gemini | PyMuPDF | openpyxl | JavaScript UI

- Built an AI grading system for assignment ZIPs with rubric generation, answer-key support, roster matching, and styled Excel reports.
- Added dual-similarity plagiarism checks, chunked grading for long submissions, viva question generation, retries, and test coverage.

ClarityHire

github.com/omerfarooq223/Clarity-Hire

FastAPI | Celery | FairLearn | React | Responsible AI | Blind Screening

- Production-grade, human-in-the-loop recruitment platform that makes resume screening more transparent, explainable, and fair.
- Anonymizes candidate information, parses resumes and job descriptions, ranks candidates with a hybrid semantic and feature-based rubric, streams live agent execution, and audits synthetic demographic outcomes without using protected attributes for scoring.

SHAP-Explained Agentic IDS

github.com/omerfarooq223/shap-agentic-ids

Python | Flask | React | SHAP | LangGraph | Security ML

- Built a hybrid intrusion detection platform combining ML detection, SHAP explanations, LangGraph reasoning, and SOC-style alerting.
- Added role-aware dashboards, forensic chat, threat-intelligence context, and autonomous red-team stress testing concepts.

FireWatch AI

github.com/omerfarooq223/Agentic-Fire-Detection

Computer Vision | YOLO-seg | Agentic RAG | LangChain

- Trained a YOLOv8-seg model on 95,000+ images for real-time fire and smoke instance segmentation, achieving 92% segmentation accuracy.
- Integrated an autonomous incident response loop in FastAPI that triggers instant safety protocols and alerts upon threshold breach.
- Built a safety-critical RAG assistant using FAISS and LangChain to retrieve context-aware hazard guidelines during live telemetry.

Additional open-source agents, computer vision models, and automation tools viewable at omerfarooq223.github.io | github.com/omerfarooq223.

TECHNICAL SKILLS

AI & ML: LLM Integration (Groq, Gemini, LLaMA) | Agentic AI Architecture | Multi-agent Patterns | Machine Learning | Computer Vision (OpenCV, YOLO) | NLP | SHAP | Scikit-learn

Languages & Frameworks: Python (Advanced) | C++ | SQL | HTML/CSS/JavaScript | FastAPI | Flask | React/Vite

Tools: n8n Workflow Automation | Git/GitHub | PyMuPDF | openpyxl | python-docx | Jupyter | Docker | Postman | VS Code

ADDITIONAL

Languages: Urdu (Native) | English (Advanced) | Turkish (Basic)

Interests: Competitive programming | Open-source AI projects | Agentic AI development | Applied cybersecurity

Certifications: DataCamp (AI Engineer for Developers Associate, OpenAI API) | Anthropic | OpenAI Academy | Google